

Alvi Ahmmed Nabil

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[Website](#) · [LinkedIn](#) · [Hugging Face](#) · [GitHub](#)

RESEARCH INTERESTS

GPU computing · On-device AI · Human-AI interaction · Model optimization · Federated learning

My goal is to make AI models efficient enough to run on the hardware people already own.

EDUCATION

Khulna University of Engineering & Technology (KUET) January 2018 – March 2023

BSc in Computer Science & Engineering · CGPA 3.55 / 4.00 (3.82 average in the last four semesters) ·

Khulna, Bangladesh

- Undergraduate thesis in natural language processing, published as first author at IEEE EICT 2023.
- Solved 500+ problems across LeetCode, Codeforces and UVa through competitive programming.

Government Syed Hatem Ali College 2017

Higher Secondary Certificate, Science · GPA 5.00 / 5.00 · Barishal, Bangladesh

Barishal Zilla School 2015

Secondary School Certificate, Science · GPA 5.00 / 5.00 · Barishal, Bangladesh

PUBLICATIONS

Bangla Emergency Post Classification on Social Media Using Transformer-Based BERT Models

Alvi Ahmmed Nabil, Dola Das, Md. Shahidul Salim, Shamsul Arifeen, H. M. Abdul Fattah. 2023 6th International Conference on Electrical Information and Communication Technology (EICT), Khulna, Bangladesh, 2023, pp. 1-6.

EXPERIENCE

Senior Software Engineer, bKash Ltd. April 2023 – present

Dhaka, Bangladesh

- Build and ship features on the app team for a mobile financial service serving over 80 million users: P2P Send Money, Savings, Remittance, Mobile Recharge, Merchant Payment and NFC Payment.
- Worked with the advanced research team on a debit and credit card scanner: the app captures a card image, transmits it encrypted for OCR, and returns extracted fields in real time to remove manual entry from wallet onboarding.
- Kept that round trip fast and reliable across millions of onboarding requests, which is where I first saw what it costs to serve a model in production.
- Overhauled legacy Dart modules into a modern Clean Architecture standard, untangling UI from data layers to make the app more resilient and easier to scale.

SELECTED PROJECTS

(Thesis) **Bangla Emergency Post Classification** 2023

- A nine-class emergency triage system for Bangla social media, trained on 5,836 posts I collected and annotated. Using Python, PyTorch, Transformers, XLM-RoBERTa, Streamlit. ([GitHub](#))

VisualML 2026

- An iOS app that draws every intermediate state of a machine learning pipeline, so the middle stops being a black box. Using Swift, SwiftUI, Python, NumPy, Accelerate. ([GitHub](#))

- Scan a bank card, run OCR on a backend service, and get structured editable fields back, stored encrypted on the device. Using Flutter, Dart, FastAPI, PaddleOCR, AES-256-GCM. ([GitHub](#))

TEST SCORES

- **GRE:** Quantitative 167 · Verbal 155
- **IELTS:** Overall 8.0 (Listening 8.5 · Reading 9.0 · Writing 7.0 · Speaking 6.5)

TECHNICAL SKILLS

- **Programming Languages:** Python, Dart, Swift, C, C++, Java, SQL
- **Markup & Database:** HTML, CSS, MySQL
- **Machine Learning:** PyTorch, TensorFlow, Transformers, NumPy, scikit-learn
- **Mobile & Backend:** Flutter, SwiftUI, UIKit, FastAPI, Firebase
- **Document Processing:** LaTeX, Microsoft Word, PowerPoint, Excel
- **Version Control & Tooling:** Git, GitHub, Docker

ACTIVITIES

- **HACK (Hardware Acceleration Club), KUET.** Batch representative.
- **BitFest 2019, KUET.** Organizer and volunteer.
- **Competitive programming.** 500+ problems solved across LeetCode, Codeforces and UVA.

AWARDS

Dean's List, Faculty of Electrical and Electronic Engineering, KUET

- Outstanding academic performance, average GPA 3.91 / 4.00; Fourth Year Examination, Academic Session 2020–2021.

Bangladesh Mathematical Olympiad

- Regional Champion, 2014, Barishal, Bangladesh; Level: Secondary
- National Runner-Up, 2012, Bangladesh; Level: Junior
- Regional Champion, 2012, Barishal, Bangladesh; Level: Junior
- National Runner-Up, 2009, Bangladesh; Level: Primary
- Regional Champion, 2009, Barishal, Bangladesh; Level: Primary

Bangladesh Astronomical Olympiad

- First Runner-Up, 2014, Bangladesh; Level: Secondary